Environmental Data Analysis in R (GEOG 4/5/7 9073)
Kent State University
College of Arts and Sciences
Department of Geography
Spring 2026 (SIS ID: 13345.202610)

Class meetings
Room: 442 McGilvrey Hall
Meetings: TuTh 9:15-10:30am

Instructor
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Department Chair
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Course description
This course aims to teach students basic concepts, skills, and tools for working with data and open source tools grounded in the R programming language. This is a project-based, student-led course that explores how to implement GIScience concepts, theories, and methods using R and R-based tools. Students will develop algorithms and programs to edit, query, manipulate, visualize and analyze spatial data. Students will also generate graphical output (maps and other plots) and create reproducible workflows. Students will develop a public-facing portfolio documenting their learning and course outputs to communicate their skills to potential mentors, employers, and the public.

Learning objectives
By the end of the term, students will be able to successfully:
- Demonstrate a familiarity with the R programming language in the context of geospatial analysis
- Write self-contained functions to automate geospatial tasks
- Analyze model workflows and describe computer code and algorithms in plain language
- Create small-scale programs that interface with web-based tools
- Practice good programming practices
- Plan, develop, and execute a programmatic analysis of a dataset

Prerequisites
BSCI 40224 or GEOG 39002 or MATH 10041

Required materials

Brunsdon, C., Comber, L. 2025. An Introduction to R for Spatial Analysis and Mapping (Spatial Analytics and GIS). Third Edition. SAGE.

Suggested materials

Wickham, H. 2017. R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media. (if you are unfamiliar with R or other programming language)

Course policies

Class format

Primary instruction will take place in a format that combines traditional lectures with in-class activities. Lab assignments will provide students with the opportunity to demonstrate their work in a practical setting.

Late work

Unless otherwise noted:

- all assignments are due on the specified due date
- late items will be accepted, but will be penalized 20% of the potential points for each day that they are late.

Changes to the syllabus

Any changes to the syllabus will be communicated via email and posted on the Canvas course page.

Working in the Lab

You are free to work in the lab whenever it is open and there is not another class using the lab. Be respectful of other classes. Students have access to the room when the building is open.

Digital file storage on lab computers should always be considered temporary. Students are encouraged to save each lab in a separate directory and sync their data with GitHub. ***YOU ARE ALWAYS RESPONSIBLE FOR YOUR DATA!*** Make backups as necessary.

Collaboration
In this class, students are not allowed to collaborate with others on any exam. Do not share your work with others or ask others to see their work.

While you may choose to interact with other students on laboratory assignments and the final project, all submitted work is expected to be your own. All write-ups, discussion statements, and other work should be your own, individual thoughts. Your maps and project should also reflect work that is independent and unique to you.

Students who do not follow these policies will be reported to the College for academic dishonesty. If you have questions regarding this policy, it is your responsibility to ask them.

Your Responsibilities
You have a responsibility to help create a classroom environment where all may learn. At the most basic level, this means you will respect the other members of the class and the instructor and you will treat them with the courtesy you expect to receive in return. This policy applies to all forms of communication in this course. Any email correspondence will be conducted via your Kent State email address.

Miscellany
Be honest and have integrity in your work. For example, do not increase the perceived length of a lab report by increasing the size of punctuation or manipulating spacing. Be kind and be polite. Finally, you will get out of this class what you put into it – be prepared, participate, and be attentive, and you will be successful.

Preparing for lab and lecture

Using the GIS Lab and Software
We will use RStudio to write and organize our work. RStudio is cross-platform and is available free of charge on the web.

Other tips
- Read relevant materials before the lecture.
- If there are topics you would like to hear more about, please bring share your ideas to make the class more relatable and interesting.
- Read the entire lab document *before* you attempt the lab assignment.
- Take advantage of office hours.
- Post questions to the appropriate Canvas forum.
- Take breaks.
- Do not leave assignments until the last minute.

Assessment

Lab assignments (60%)

There are 6 required lab assignments. Labs will become more complicated and include less guided instruction as the semester progresses. Each lab assignment will have a corresponding Canvas Discussion Board where you can post (and respond) to questions. ***Please use the discussion board to attempt and answer your questions about lab assignments prior to emailing your instructor.*** Using the discussion board will ensure that all students will benefit from the questions and their responses. You may work on labs at any time. A weekly schedule for the lab will be posted on Canvas. Feel free to work on labs elsewhere if you have access to the necessary software and are comfortable working independently. Lab assignments are to be completed at your own pace, but they must be submitted prior to the due date. The due date for each assignment is included in the instructions.

All completed lab assignments must be submitted to the course Canvas site. No other submissions will be accepted.

Final Project (35%)

The final project is a chance for you to apply your programming, scripting, and automation skills to an area of your interest. The final project is a required component of the course. You must submit this assignment to pass the course regardless of your other assignments such as exams and labs. Additional documents explicitly defining the guidelines and milestones of the project, and example projects will be shared. Final project has four milestones: 1) a project proposal that includes preliminary data, 2) an in-class presentation that will update the class on your progress and serve as opportunity to receive feedback, 3) a final in-class presentation on your term project, and 4) a project report, relevant code, and any relevant analysis. There will be a series of project workshops throughout the semester to provide you with the rubric and expectations for the final project proposal, data, and final report deliverables.

Participation (5%)

You are expected to engage in in-class discussion with your peers and instructor.

All assignments should be submitted to the corresponding Canvas assignment before the due date.

Evaluation scale

Grade	% of Points	Grade	% of Points	Grade	% of Points	Grade	% of Points
A	94-100	B+	87-89	C+	77-79	D+	67-69
A-	90-93	B	84-86	C	74-76	D	64-66
		B-	80-83	C-	70-73	D-	60-63
						F	Below 60

Grades will be based on the following:

Assessment	**Total points**
Lab assignments (6 x 100 points/each)	500
Participation	50
Final project	*350 total*
Proposal	50
In-class update	20
Final presentation	80
Final paper	200
Total	**1000**

Extra credit

Extra credit assignments and opportunities will not be offered.

Graduate students

For those graduate students enrolled in the course, the requirements of the final project will be expanded to include: 1) an additional 3-4 pages in your report, 2) code documentation, and 3) an additional 5 minutes in your final presentation to the class. You are also required to produce a cover page for your GitHub page/portfolio

Tentative Course Schedule

Week	Topic
1	Introduction, introduction to R and RStudio, GIScience refresher
2	Data structures and plots, the Tidyverse
3	Basics of spatial data handling in R, VCS
4	Writing functions and writing reproducible code
5	ESDA and R as a GIS
6	Spatial relations and data operations
7	Point pattern analysis
8	Localized spatial analysis
9	**Spring Break**
10	**AAG Week**
11	Rasters
12	Time series and hierarchical models
13	Mapping with R
14	Interactive mapping
15	Applications: topics TBD (based on student interests)
16	Final project presentations
Finals	**Report due**

University policies

Request for Religious Accommodations
The University welcomes individuals from all different faiths, philosophies, religious traditions, and other systems of belief, and supports their respective practices. In compliance with University policy and the Ohio Revised Code, the University permits students to request class absences for up to three (3) days, per semester, in order to participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization. Students will not be penalized as a result of any of these excused absences.

The request for excusal must be made, in writing, during the first fourteen (14) days of the semester and include the date(s) of each proposed absence or request for alternative religious accommodation. The request must clearly state that the proposed absence is to participate in religious activities. The request must also provide the particular accommodation(s) you desire.

You will be notified by me if your request is approved, or, if it is approved with modification. I will work with you in an effort to arrange a mutually agreeable alternative arrangement. For more information regarding this Policy you may contact the Student Ombuds (ombuds@kent.edu).

Course registration and withdrawal
The official registration deadline for this course is January 18th. University policy requires all students to be officially registered in each class they are attending. Students who are not officially registered for a course by published deadlines should not be attending classes and will not receive credit or a grade for the course. Each student must confirm enrollment by checking his/her class schedule (using Student Tools in FlashLine) prior to the deadline indicated. Registration errors must be corrected prior to the deadline. The course withdrawal deadline is March 29th.

Student Accessibility Services Statement
Kent State University is committed to inclusive and accessible education experiences for all students. University Policy 3342-3-01.3 requires that students with disabilities be provided reasonable accommodations to ensure equal access to course content. Students with disabilities are encouraged to connect with Student Accessibility Services as early as possible to establish accommodations. If you anticipate or experience academic barriers based on a disability (including mental health, chronic medical conditions, or injuries), please let me know immediately.

Basic Needs Support & Mental Health Well-being
Kent State University is committed to supporting the overall well-being of our students. This support may take the form of assisting students with basic needs such as food and housing. We recognize that the absence of secure housing and access to food makes it difficult for students to achieve their best in and out of the classroom. If you, or someone you know, are unable to afford groceries or lack a safe, secure, and reliable place to live, please visit the

CARES Center basic needs resource website: https://www.kent.edu/carescenter/basic-needs-resources.

Mental health challenges may also make it difficult for students to reach their full potential. KSU's mental health and wellness resource page provides information on education and awareness, mental health services, and advocacy intervention. To learn more, please visit Kent State's mental health resources and support website at https://www.kent.edu/mentalhealth.

Academic honesty
The Administrative Policy regarding student cheating and plagiarism (3-01.8)has been revised to include details on generative AI (GAI). The policy defines GAI as any internet-based generative artificial intelligence program that makes use of large language model algorithms to create content and that the use of GAI to generate content for assigned coursework, **unless expressly permitted by the instructor** in the syllabus or assignment instructions, is considered to be cheating. https://www.kent.edu/policyreg/administrative-policy-regarding-student-cheating-and-plagiarism.

Disruptive student policy
I expect all people in this academic space will be respectful. If a person is disruptive, I will enforce university policy.

Examples of disruptive behavior may include but are not limited to:

- Monopolizing class time with excessive comments or questions
- Interrupting class sessions or assignments
- Abusing faculty office hours
- Repeatedly speaking in an elevated or angry/aggressive tone
- Attacking the beliefs or conclusions others draw in class.

Please see KSU policy 4-02.2 and process for disruptive students, which include referral to the Office of Student Conduct as I deem necessary.

Registration and records policy
Only students who have been formally admitted to Kent State University may register for coursework and pay the appropriate fees. An official registration is a record of the students' approved schedule or classes maintained online in the university's student information system. Students who are not officially registered for a course by published university deadlines should not attend classes and will not receive credit or a grade for the course.

Kent Campus Academic Support Statement
Kent State recognizes many students face challenges and we are committed to supporting your academic journey when you need help. Please check out these resources to help as you build your support system:

- What is the first step I should take to get academic support for this class?
  - Reach out to your instructor!
- Where can I get help from another student who earned a good grade in this class?

    - **Tutoring**
- Where can I go if I need assistance with how to study and meet my academic goals?
    - **Academic Coaching**
- Who can review my writing and help me properly cite my work?
    - **Writing Commons**
- Where should I go when I don't know where to go?
    - **Academic Advising**
    - **TRIO Student Support Services**

There may be additional resources, just ask